SCENE-LEVEL TYPE-CONSISTENT RELABELLING FOR TRANSPARENT GLASS DETECTION

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ABSTRACT

GlassNICOL, a transparent-glass detection dataset, types each glass automatically from a depth-derived height in every view. Type is constant, so adjacent views should agree; the released training labels often disagree, while the hand-labelled test splits almost never do. We repair the labels with the structure that produced them. A scene is static, so every view of a glass views one point on the table: we register views through that plane, carry each box by its base point, chain the views of each glass, and give every box in a chain its most frequent type. Detectors trained on the relabelled set, five (accelerator, seed) cells under a sealed protocol, gain out of distribution in type accuracy and six-class average precision over the raw labels and over an adjacent-frame vote; on the standard split six-class average precision does not improve and recall falls about two points. Only type ids change.

Index Terms— transparent object detection, label noise, multi-view consistency, relabelling, glass detection

1. INTRODUCTION

Transparent objects are expensive to label: depth sensors return little through glass, so ground truth comes from a proxy the sensor can see. ClearGrasp swaps each object for a spray-painted copy [1]; KeyPose labels keypoints in a few views and propagates them through known camera poses [2]; GlassNICOL photographs each tabletop scene in 25 views and records it twice more, once with each glass capped to measure its height and once with a chalk-sprayed copy in its place for depth; that height, matched to the nearest known glass, is the type [3]. KeyPose checks its geometry across views; nothing checks that the type assigned in one view agrees with the next.

We check. After registering adjacent training views and matching boxes, 18.1% of matched box pairs that are the same physical glass carry a different type; the hand-labelled test splits pass the same test at 0.0% and 0.3% (Section 2.1).

The structure that produced the labels can also repair them, and the scene says how. A GlassNICOL scene is static: the glasses stand on the table and only the head camera moves. Every view of a glass is therefore a view of one point of the table plane, its base, and a homography of that plane carries points of it exactly from one view to another. We register views pairwise through the table plane, carry each box by its base point, link boxes whose base points coincide, and merge the links into one chain per glass across the scene; each chain gives its most frequent type to every box in it (Fig. 1, Section 2). Our earlier baseline linked only adjacent frames by box overlap, and such chains break at a missed frame; registering through the table plane and carrying base points over two hops raises the median votes per box from 5.5 to 10 while merging no two glasses of different type on the manual splits. 12.3% of type ids change and no box moves.

Contributions: (i) a measurement of type-label inconsistency in a released benchmark against its own manual test labels; (ii) a scene-level relabelling method that uses only the dataset's own views, associates through the support plane without camera poses or depth, and changes type ids alone; (iii) evidence over five (accelerator, seed) cells on two accelerators under a protocol fixed before training (Section 3).

Relation to prior work. Repeat-factor sampling [4] and data pruning [5, 6] choose which images the network sees and take every label as given; we change what a label says. Label noise is usually found from model confidence [7, 8] or the training loss [9], or absorbed by robust training [10, 11]; we locate it from the capture geometry, without a model. Tubelet methods smooth a detector's own output at test time, by re-scoring [12, 13] or propagation [14], and offboard auto-labelling turns detections over a sequence into training labels [15]; we vote on released labels, not on detections. Closest to us, a pseudo-tracklet majority corrects a recogniser's class over a track [16]: that vote is over detections, ours over released training labels, and the capture geometry, not a detector, groups the views. Ground-plane homographies associate people across cameras by their feet [17]; we use the table plane the same way, across views of one camera. Label Fusion reprojects one 3D label into every frame [18], whereas GlassNICOL types each view separately and the disagreement enters there; KeyPose discards scans whose keypoints disagree [2], whereas we keep the frame and correct the label.

2. PROPOSED METHOD

2.1. Type labels disagree on the same object

GlassNICOL assigns type automatically (we call this step the labeller): the height of a depth cluster above the table plane, measured in the capped pass, is matched to the nearest of six known glass heights (5, 9, 11, 13, 17.5 and 22 cm) within 3 cm, all constants of the released labelling code [3]. Height is measured per view, but type is constant, so one glass should carry one type across the 25 views of its scene. We register each of the 2,280 consecutive training frame pairs by background features, carry every box through the registration and match it to the box it overlaps in $f+1$ at intersection over union (IoU) above 0.5; pairs whose registration fails are matched unwarped. Of the 2,775 matched pairs, 502 (18.1%) carry a different type (Fig. 2), 87% of them between glasses of neighbouring height.

The dataset authors likewise use manual test annotations to reduce noise from automatic labelling [3]. Those splits yield 0 flips in 93 matched pairs on the standard test, at most 1 in 316 (0.3%) on the out-of-distribution (OOD) test. The labeller also places two boxes on one glass in 51 training frames, 44 of 53 such pairs with different types, and none in the manual splits.

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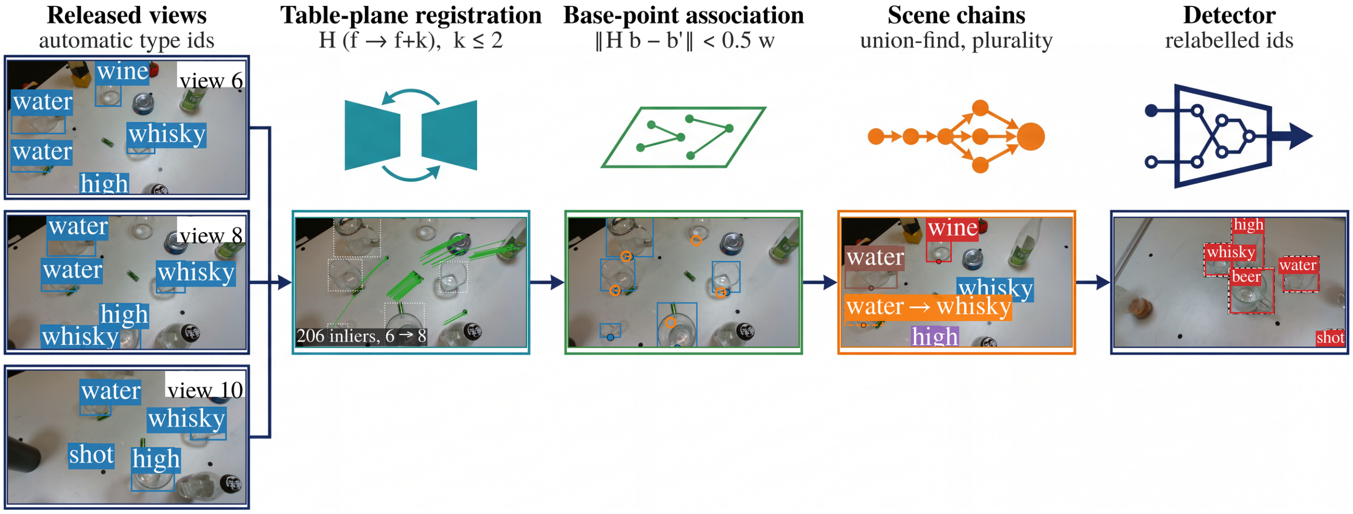

Fig. 1. Overview. Views of a static scene are registered through the table plane, each box is carried by its base point and linked to the box it lands on, union-find merges the links into one chain per glass, and the chain's most frequent type replaces every type id in it. Photographs: views 6, 8 and 10 of one training scene with released boxes and types; inliers of the view 6 to 8 homography as displacement vectors; base points of view 6 carried into view 8 (open) and linked to its own (filled); view 6 with one colour per chain and a dashed box where the vote replaced the type; a scene-vote detector on a manual OOD test frame, manual boxes dashed, hand-picked where every manual box is detected with the correct type (Fig. 3 chooses its frames by rule). Overlays are real; stage symbols schematic.

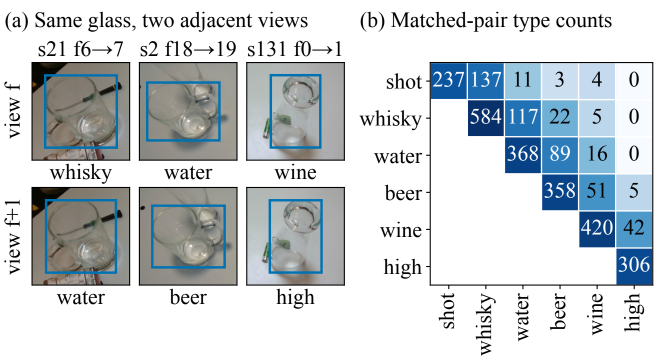

Fig. 2. Automatic type labels are not constant on one object. (a) The same glass in adjacent training views f and $f{+}1$, cropped from released frames with the released box and type. (b) Released types of the 2,775 box pairs matched across adjacent views after background registration, each unordered pair counted once; 502 (18.1%) disagree, 87% of them between neighbouring heights. Manual test labels flip 0/93 and 1/316.

The misreads mostly join neighbouring heights and mostly run downward. Treating the views of one glass on an adjacent-frame chain as repeated readings and estimating the labeller's confusion by expectation–maximisation [19], every type but the shortest is read low more often than high: a water glass in 18.5% of views against 4.7%, a wine glass 11.0% against 0.8%, and only the shot glass, which has no shorter type, is misread upward, as whisky in 8.9%. The full estimated confusion is released with the code.

2.2. Association through the table plane

Frames f and $f{+}k$, $k \leq 2$, are registered by background features: the label boxes are dilated $1.2\times$ and masked out, up to 1,500 ORB features [20] are extracted from what remains and matched under a 0.75 ratio test, and a homography is fitted by RANSAC (random sample consensus) [21] at a 3-pixel threshold, kept when at least 30 matches are inliers and they are half of the matches or more. The background is the table, so the homography is that plane's and carries points on it exactly. A glass's base point is the bottom-centre of its box, an approximation the tolerance absorbs; the rest of the box moves with parallax. The release annotates base points for the training split only, not for the hand-labelled splits where we check the association, so we read them off the boxes. Each box of f is carried into $f{+}k$ by its base point and linked, one-to-one by distance, to the box whose base point lies within 0.5 of the mean box width. Union-find merges links across the scene into chains, one glass seen in several views, under one constraint: a chain holds at most one box per frame, and a link violating it is refused (64 of the links the registered pairs propose).

2.3. Vote, coverage and validity

Each chain takes its most frequent type; ties are left unchanged. Only the type id changes: no box moves, no image is added or removed, and roster and hyperparameters match the raw-label arm of Section 3. The association runs on all 95 training scenes, of which the detector roster below uses 70: 4,097 of 4,465 frame pairs register, and 4,336 boxes fall into 786 chains with a median of 10 views per box, 91% voted on by three views or more, 535 (12.3%) changing type and 140 in tied chains keeping their label. Our earlier adjacent-frame overlap vote, which registered 1,748 of 2,280 pairs by a partial-affine warp, gives 1,561 chains, 878 of them a single box, a median of 5.5 votes per box, and changes 364 boxes (8.4%). Ablating the ingredients under the homography registration (post hoc; full table with the code): overlap linking gives a median of 8 votes over one hop and 9 over two, the base point over one hop 9 votes but 2 disagreements on the OOD split, and both together 10 votes at 12.3% of ids changed. Three of the four settings merge no two glasses of different type on either manual split, and among those three ours votes with the most views. That baseline is the tracklet arm below. The vote also runs on the validation split, but selection scores generic average precision (AP),

which collapses the six types, so validation type ids never enter it.
Validity. Manual types are constant per object, so a within-chain type disagreement on the hand-labelled splits is a merged pair of glasses. The association above produces 0 of 167 and 0 of 351 disagreeing boxes on the two splits, with median chain lengths 10 and 15, and 163 of 167 and 346 of 351 boxes in chains of three views or more. Hops and tolerance were chosen by this check, on a grid of two to four hops and four tolerances, before any scene-vote detector was trained, taking the loosest setting that disagrees nowhere; the zero above is in-sample, and at that tolerance three and four hops leave four disagreements on the standard split. The manual splits never enter training.
Scope. A vote over views removes disagreement that varies by view and inherits whatever is constant across views, such as the class-share difference of Section 4.

3. EXPERIMENTAL RESULTS

3.1. Setup

Data. The official GlassNICOL head-camera split [3]: 2,375 training images from 95 scenes, 125 validation images, and hand-labelled standard and OOD tests (66 and 75 images, 3 scenes each). The roster, fixed beforehand, uses 70 scenes (1,750 images).
Detector and regime. YOLO11n [22] (2.59M parameters [23]), 640×640 inputs, batch 64, AdamW [24] at $\mathrm{lr}_0 = 10^{-3}$, cosine decay over 300 epochs. All arms train under the same repeated-exposure regime: the 1,538 training images whose transparent and chalk-coated captures register by the same background matching appear twice, 3,288 slots and 15,600 updates, and the arms differ only in training type ids. Seeds 1337, 3407 and 4567 run on accelerator A (an H200 GPU) and seeds 1337 and 5678 on accelerator B (an RTX PRO 6000 GPU); a cell is one (accelerator, seed) set of the three arms, an arm being one label variant on the shared roster. On A the raw and scene-vote runs of a cell share a GPU and the tracklet run used another; on B all share one.
Scoring. The checkpoint with the best validation generic AP is scored at confidence floor 0.001 with non-maximum suppression at IoU 0.7: COCO (Common Objects in Context) box AP over IoU 0.50–0.95 [25]; generic AP collapses the six types to one. Within each image, GT boxes in ascending annotation-id order greedily take the unused prediction of maximum IoU (≥ 0.5, confidence ≥ 0.25), irrespective of type. Pairwise type accuracy is the correctly typed fraction on boxes both compared arms localise; recall is the localised fraction per arm.
Existing methods compared. Two standard answers to label noise are trained as arms under this protocol. Label smoothing [26] keeps the released ids and smooths the classification target of assigned anchors by 0.1. Detector-confidence relabelling replaces a type id when a detector disagrees with confidence at least 0.5, judged out of fold by a model trained on the scene-disjoint half of the roster, a judge seeing half the scenes at a quarter of the updates; it is the confident-learning idea [7, 27] at one global threshold, and changes 22.5% of the ids on that roster against our 11.7%.
Protocol. The comparison was preregistered before any scene-vote detector was trained: the three arms, the five cells, the checkpoint rule, the four metrics above, and four predictions, a positive mean type-accuracy gain over the raw labels on both splits (P1), an OOD gain exceeding the tracklet vote's (P2), recall and generic AP within two points (P3), and a positive OOD gain in at least 4 of 5 cells (P4). The sealed protocol, every evaluation receipt and the code are at `https://github.com/HeunSeungLim/glassnicol-scene-relabel`. Every contrast is read as a mean over cells with the count of positive cells beside it [28].

3.2. What the labels change

Table 1 pools the cells; the per-cell table is released with the code. Two contrasts hold across cells. Out of distribution, six-class AP rises +7.1 points over the raw labels in 5 of 5 (95% t-interval [+1.8, +12.3]; +6.4 on A, +8.0 on B), and +6.4 over the tracklet vote ([+0.6, +12.2]), which itself did not move (+0.7). On the standard split localisation recall falls in every cell, by 2.0 points on average ([-3.7, -0.3]). The type-accuracy gains are larger but less certain: out of distribution the vote gains +9.2 points in 4 of 5, with paired bootstrap intervals over 2,000 image resamples above zero in 3 cells, touching it in 1 and below in 1; resampling scenes leaves 8 of 10 per-cell intervals over both splits including zero, and the across-cell interval is [-5.7, +24.1]; over the tracklet vote it gains +4.0 (4 of 5). On the standard split type accuracy changes by +0.8 over the raw labels (3 of 5 cells positive) and -1.8 over the tracklet vote (1); on the shared denominator of Table 1 the first contrast is 84.1 against 84.2, so the two box sets give opposite signs to a small difference. Six-class and generic AP change by -1.9 and -1.7 (2 and 1 of 5 positive). The standard six-class change also splits by accelerator, -7.0 on A ([-23.1, +9.1]) and +5.8 on B ([-2.7, +14.3]), against +6.4 ([-7.8, +20.7]) and +8.0 ([-9.8, +25.8]) out of distribution; with three cells against two and one shared seed, accelerator and seed are not separable.
Preregistered predictions. P1 held: the mean type-accuracy gain over the raw labels is positive on both splits, +0.8 and +9.2. P2 held: out of distribution the gain exceeds the tracklet vote's +5.0 (pairwise contrasts use different box sets and need not add). P4 held: 4 of 5 OOD cells are positive. P3 did not: standard-split recall falls 2.0 points, just past the bound, and generic AP 1.7; out of distribution both stay within it. Scoring the last epoch instead (post hoc) gives type accuracy -0.6 on the standard split, so P1 holds only under the preregistered rule, and +11.3 out of distribution; recall moves +0.5 and -1.7, so the OOD gain holds under both rules while the standard-split recall loss does not. Under that rule the tracklet vote also gains out of distribution, six-class AP +7.0 in 5 of 5 cells, but the scene vote still leads it by +4.1 points in 5 of 5 ([-0.2, +8.3], against [+0.6, +12.2] under the preregistered rule): the lead survives the rule change, but its interval now includes zero.

3.3. Comparison with existing methods

Table 1 sets the two standard answers to label noise, and our earlier vote, beside ours; Fig. 3 shows the five detectors on manual test frames. Label smoothing does not change out-of-distribution accuracy (-0.5, above the released labels in 3 of 5): absorbing the noise does not recover a wrong type id. Detector confidence changes the most labels and loses the most accuracy, -15.5 out of distribution, above the released labels in 0 of 5. It rewrites 22.5% of the ids against our 11.7%, and 81.8% of those differ from the type the geometry votes for, 48.2 points of that forced by the edit counts alone: a box it moves and we leave alone differs by construction. Under the preregistered rule the scene vote alone gains out of distribution in every cell, +7.1 points in 5 of 5, against +0.7 in 2 for the adjacent-frame vote. At threshold 0.8 (post hoc, A) it edits 12.8% of the ids, about as many as our vote, and still falls 11.2 points out of distribution in all 3 cells while beating us on the standard split: fewer edits soften the loss without making it a gain. Smoothing is the better choice on the standard split, ahead of us in every cell, so the case for relabelling is out of distribution. The last epoch keeps the order: +0.7 for smooth-

Table 1. The five training-label variants on the official test splits (manual labels), mean over the 5 cells, all in percent. Δ is the change from the released labels, before rounding, with the cells above them. Six: six-class AP; Gen: generic AP, the six types collapsed to one; Rec: localisation recall; Type: type accuracy. Every row trains the same detector on the same roster, budget and seeds, three cells on A and two on B, differing only in the training type ids, or for smoothing in the classification target. Type is scored on the boxes all five arms localise in a cell (121–135 of 167 on the standard split, 222–273 of 351 out of distribution), one denominator for the five rows; the preregistered predictions use pairwise common sets instead. Entries are rounded, so subtracting two need not reproduce a contrast in the text; Type differs further because those contrasts use pairwise box sets. Label smoothing and detector confidence are post hoc, added after the sealed protocol. In the four metric columns of each split, ranked before rounding, best bold and second blue.

	Standard					OOD				
Training labels	Six	Δ	Gen	Rec	Type	Six	Δ	Gen	Rec	Type
Released labels	65.4	—	84.2	90.8	84.2	53.7	—	66.6	79.7	76.0
Label smoothing	**68.7**	+3.3 (3/5)	84.2	**91.3**	84.7	53.1	−0.5 (3/5)	65.6	79.9	76.4
Detector confidence	62.2	−3.1 (1/5)	83.9	88.3	81.6	38.2	−15.5 (0/5)	65.4	79.0	62.9
Adjacent-frame vote (earlier)	65.6	+0.2 (3/5)	**85.7**	89.0	**86.1**	54.3	+0.7 (2/5)	**67.1**	**80.1**	81.3
Scene vote (ours)	63.5	−1.9 (2/5)	82.4	88.7	84.1	**60.7**	+7.1 (5/5)	66.3	78.4	**84.8**

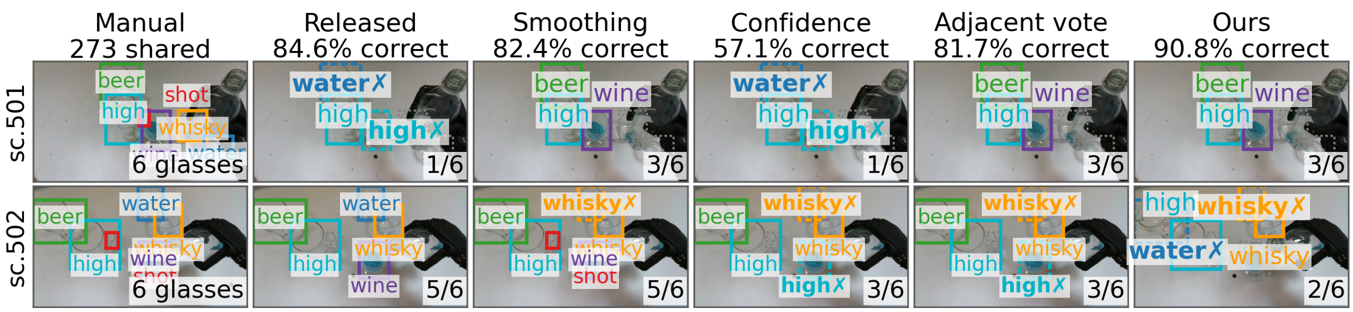

Fig. 3. The five training-label arms on OOD test frames, accelerator A seed 1337, from the frozen predictions behind Table 1: the manual annotation, then detectors trained on released labels, label smoothing, detector-confidence relabelling, the adjacent-frame vote and ours. Boxes sit at the manual location, coloured by the predicted type: solid if right, dashed with a cross if wrong, grey dotted if the arm finds nothing. Headings score all arms on the 273 boxes every arm localises; a panel count is correctly typed boxes of the frame's manual boxes. Row 1 is a frame where the released-label arm gets the most boxes wrong, mistyped or missed (four tie; first by file name); row 2 is our worst frame, the unique maximum of boxes we get wrong, again mistyped or missed, that it gets right.

ing, −4.2 for detector confidence, +7.0 for the tracklet vote, +11.1 for ours (5/5).

4. DISCUSSION AND LIMITATIONS

The vote does not bring class shares closer to the tests: the shot:whisky ratio is 0.58 in raw labels and 0.51 after it, against 0.79 and 1.83 in the tests; decoding the chains under the estimated confusion moves it no closer, and separating the labeller's share from the staging's needs a per-box height the release omits.

Six-class AP is the mean over the six types, so the standard-split −1.9 is a reallocation that nets out rather than a uniform decline: shot gains +12.4 and beer +5.7 against −14.6 for whisky and −12.3 for water, while type accuracy there is a draw. The edits are short and mostly upward: 77.8% of the 535 changed ids move one step on the height ladder and 68.4% move up. The shortest type can only move up and the tallest only down; over the 395 edits that start at neither, 62.5% still move up, undoing the downward bias the labeller's estimated confusion shows.

A confidence gate changes both together. Voting only with 3 views and a margin of 2 (post hoc, A) edits 8.6% of the ids, not 12.3%: the standard six-class change becomes 0.0 (1 of 3) against −7.0, standard-split recall turns +0.8 (3 of 3) from −2.0, and the six-class gain out of distribution falls to +1.1 (2 of 3) from +6.4, while the type-accuracy gain barely moves, +3.5 (2 of 3) against +3.9 (2). Three cells cannot separate the movements, and one seed carries most of the fall out of distribution.

The association is only as good as the table: frames without background features stay unlinked, and the 51 frames with two boxes on one glass count twice.

Seed 4567 on A loses type accuracy on both splits, by eight and ten points, while every other cell gains six or more out of distribution; the mean sits inside the seed spread, so six-class AP carries the claim, and the tracklet contrast rerun on B for seed 1337 moved by 10.9/14.3 type-accuracy points, a caution for single-cell results. Checkpoint selection by validation generic AP ($r^2 = 0.04$) is shared across arms.

5. CONCLUSION

A released benchmark can disagree with itself, measurably and without a model: GlassNICOL's automatic types differ on the same glass in 18.1% of adjacent-view pairs, where its hand labels almost never do. The geometry that produced them repairs them: a homography of the static table plane carries base points between views, union-find chains them, and a plurality sets the type, moving 12.3% of ids and no box. The relabelling helps out of distribution, not in it: under the preregistered rule ours is the only arm gaining there in every cell, +7.1 six-class AP, against no change from smoothing and 15.5 points lost by a detector-written rewrite. In distribution it costs about two points of recall and six-class AP; on accelerator A a gate removes that cost post hoc, taking most of the gain with it. A pipeline that leaves a redundancy in its captures unused can use it to fix its labels.

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